Handling Missing Values

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Possible Reasons for Missing Values

- Forgotten items
- Answers not readable
- Technical problems
- Attrition
- Non-response
- ...

Types of missing values

Missing completely at random (MCAR)

- missing values occured truely at random

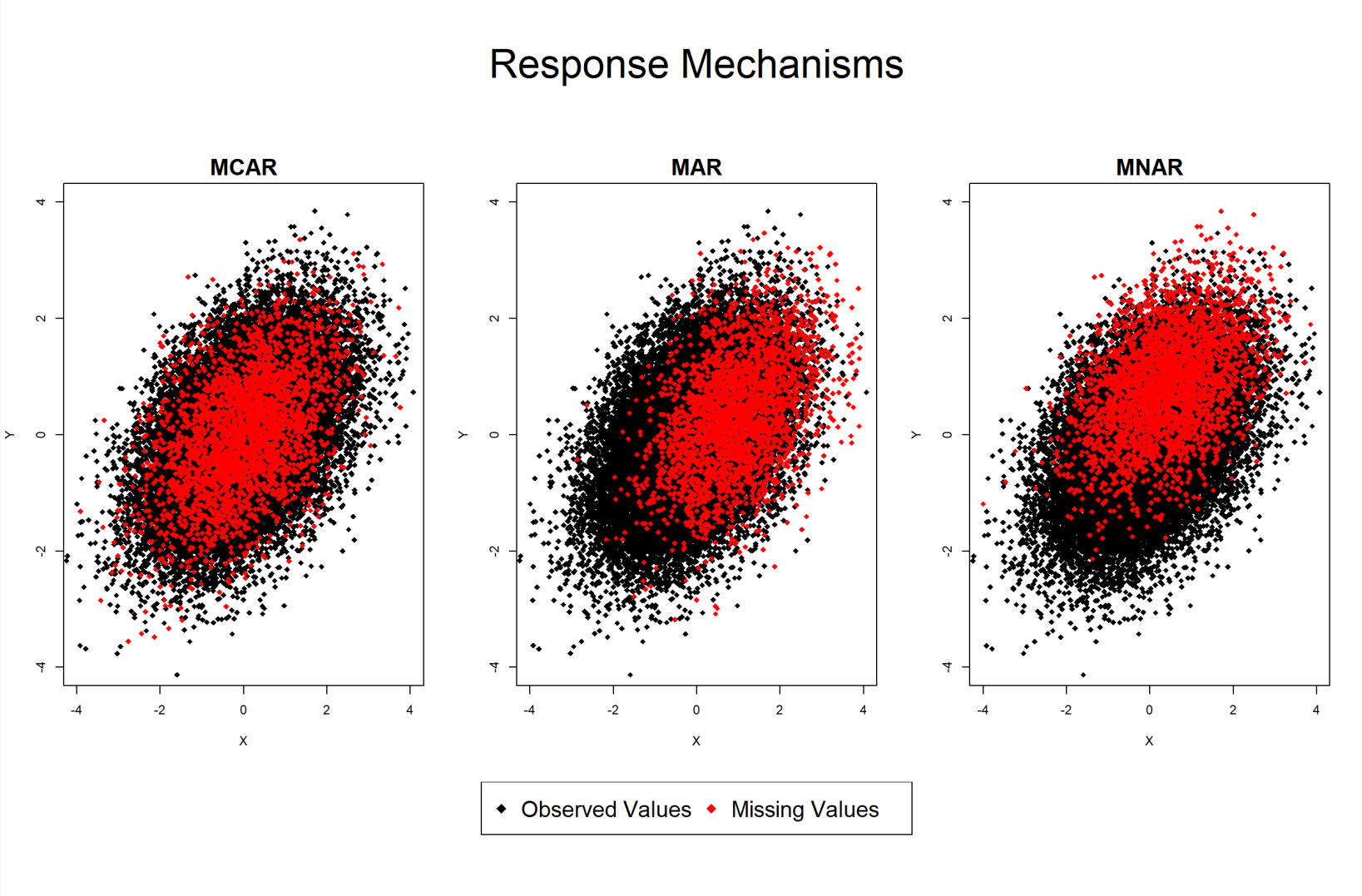
Missing at random (MAR)

- . . . actually not happening at random
- missingness is predictable using observered variables

Missing not at random (MNAR)

- missingness happening not at random and not predictable

Types of missing values: Visualization



Ways to handle missing data

Listwise exclusion

- Complete Case Analysis
- Available Case Analysis

Data based approaches

- Single Imputation
- Multiple Imputation

Model based approaches

- Full information maximum likelihood (FIML)
- Missing value estimation on the fly

Listwise Exclusion

Almost never recommended!

Complete Case Analysis:

- Exclude all observational units for which the response and/or a predictor is missing

Available Case Analysis:

- Use separate analyses for different aspects of the problem
- Analyses will be based on different data subsets

Single Imputation

Fill in missings with appropriate values

- Mean Imputation
- Last value carried forward
- Use related information from different source
- Create indicator variables for missingness
- Predict missing values with (regression) models

Single Imputation: Model Based

Fit a regression model for the variable with missingness

Deterministic regression imputation

- Use the mean predicted values
- Fails to account for uncertainty in the regression

Random regression imputation

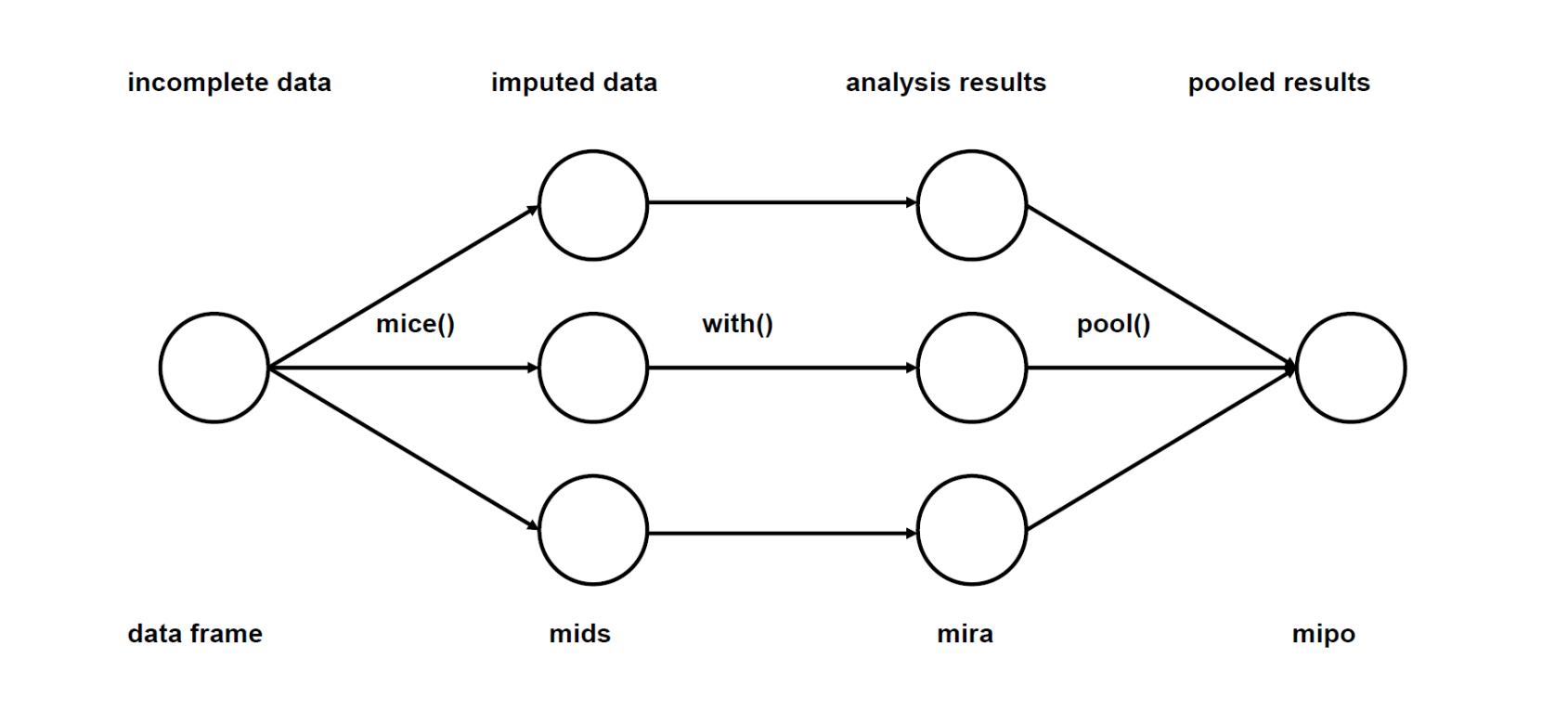
- Add the prediction error to the mean predicted values
- Accounts for uncertainty in the regression
- (Still a single imputated value)

Multiple Imputation

Necessary steps:

- Generate multiple (imputed) data sets
- Fit your statistical model to every data set separately
- Pool the results across models

Multiple Imputation with the mice Package



Multiple Imputation: Generate Data Sets

Predictive mean matching:

- Imputations are restricted to the observered values
- Can handle non-linear relations well

Linear Regression approaches:

- Very efficient approaches
- Work well for normally distributed data

Logistic / Categorical Regression approaches:

- Useful to impute categorical variables

Multiple Imputation in Multilevel Models

Impute in the wide data set:

- Imputation works as in other cases
- No explicit inclusion of the person variable necessary
- Requires transformation of all wide to long data sets

Use special imputation methods for multilevel models

- Work with the preferred format (long data set)
- Good performance if assumptions are met
- No very flexible with respect to the supported multilevel models

Model based approaches: FIML

- Use all observered data
- Include covariates which are (assumed to be) correlated with the missingness process

Advantages:

- Comes natural to multilevel models
- Requires the model to be fitted only once

Disadvantages:

- More model parameters have to be estimated
- Assessing model fit becomes more complicated
- Not applicable to all types of missing values

Model based approaches: Estimation on the fly

- Use all observed data
- Treat variables with missing values as additional response variables

Advantages:

- Extremely flexible
- Requires the model to be fitted only once

Disadvantages:

- More complex than multiple imputation methods
- Only works for continuous variables in brms

Estimation on the fly: Simple Example

Suppose:

- We are predicting `y` by `x` and `z`
- `x` contains missing values
- `y` and `x` are both normally distributed

Possible model handling missing values in `x` on the fly:

$$\eta_{xn} = b_{x0} + b_{x1} z_n$$

$$x_n \sim \mathcal{N}(\eta_{xn}, \sigma_x)$$

$$\eta_{yn} = b_{y0} + b_{y1} z_n + b_{y2} x_n$$

$$y_n \sim \mathcal{N}(\eta_{yn}, \sigma_y)$$

brms Syntax for Missing Value Imputation

Suppose variable `x` contains missings

Syntax for multiple imputation:

```
imp <- mice(dat)
brm_multiple(y ~ x + ..., data = imp)
```

Syntax for missing value estimation on the fly:

```
bform <- bf(x | mi() ~ ...) + bf(y ~ mi(x) + ...)
brm(bform, data = dat)
```

Time for exercise

'missing_values.R'

Further Reading

- Gelman, A., & Hill, J. (2007). Data analysis using regression and multilevel/hierarchical models. Cambridge University Press.
- Buuren, S., & Groothuis-Oudshoorn, K. (2011). mice: Multivariate imputation by chained equations in R. Journal of statistical software, 45(3).